Summary in Graph

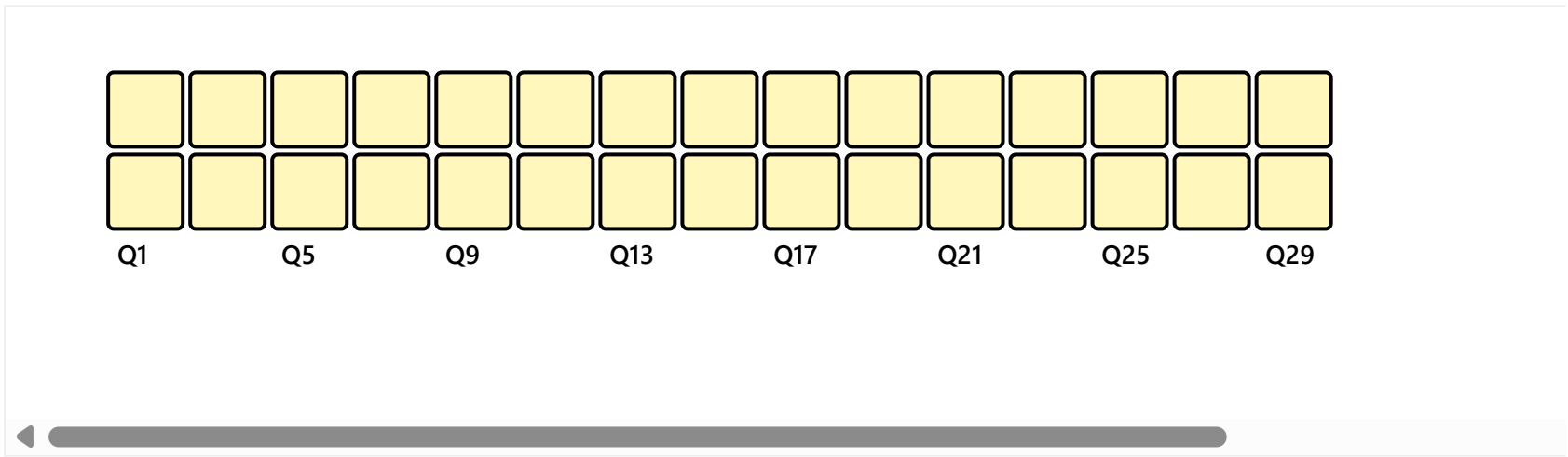
Exam Summary (GO Classes CS Test Series 2026 | Algorithms | Subject Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (15 + 15)
Total Marks:	45 (15 + 30)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (45) out of 449 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $G = (V, E)$ be an undirected connected graph. Consider the following statements:

- $S1$: The depth of a Breadth-First Search (BFS) tree starting from an arbitrary vertex $v \in V$ is equal to the diameter of graph G.

- $S2$: The depth of a Depth-First Search (DFS) tree starting from an arbitrary vertex $v \in V$ is equal to the diameter of graph G.

Which of the above statements is/are TRUE?

A. $S1$ is True and $S2$ is False

B. $S1$ is False and $S2$ is True

C. Both $S1$ and $S2$ are True

D. Both $S1$ and $S2$ are False

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

You're playing the hit new platform video game, Mega Meat Man, and are having trouble getting through Level 6006. You've decided to model the level as a directed graph, where each vertex represents a platform you can reach, and each edge represents a jump you can try to make. After extensive experimentation, you've labelled each edge with the probability (a number in $[0, 1]$) that you can successfully make the jump. Unfortunately, if you fail to make any jump, you instantly die and have to start over. Which of the following algorithms and time complexities best suit this problem?

A. Dijkstra's Algorithm

B. Bellman-ford Algorithm

C. $O((V + E) \cdot \log V)$

D. $O(V . E)$

Your Answer: | **Correct Answer: A;C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a binary string S (a string consisting only of 0 s and 1 s) of length n. What is the correct time complexity of sorting the string S?

A. $\Theta(\log n)$

B. $\Theta(n)$

C. $\Theta(n \cdot \log n)$

D. $\Theta\ (n\hat{}2)$

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following three claims:

1. $2^{(\log n)^2} = O(n^{1.5})$

2. $2^{2n+1} = O(2^n)$

3. $2^{\sqrt{n}} = O(n^{\log n})$

Which of the following claims are FALSE?

A. 1 and 3

B. 2 and 3

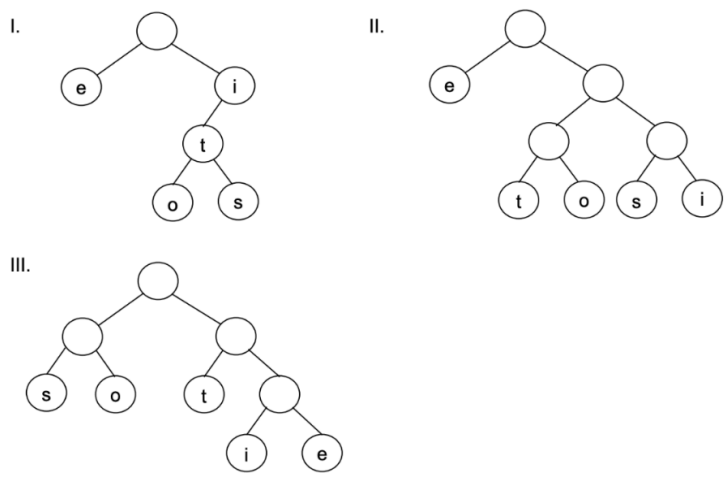
C. 1 and 2

D. 1, 2 and 3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

A huffman tree is constructed for a text document containing 5 characters. The character $\mathtt{e}$ appears most frequently, and the character $\mathtt{i}$ has the next highest frequency. Which of the following could be the huffman tree for this document?

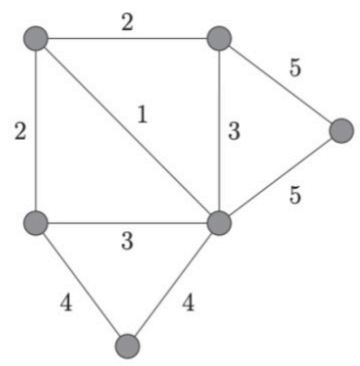


A. II
B. II and I
C. I and III
D. I, II and III

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 06sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

How many distinct minimum weight spanning trees does the following undirected weighted graph have?



A. 1

B. 2

C. 4

D. 6

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following pseudocode:

```
procedure StarDash(n)
  if n = 0 then
    print '*'
  else if n = 1 then
    print '-'
  else
    StarDash(n - 1)
    StarDash(n - 2)
end procedure
```

Which of the following is the correct output for StarDash(5)?

A. - * - - * - **

B. - * - - - * - -

C. - * - - * - * -

D. - * - - * - - -

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the recurrence relation:

$$T(n) = T(n-1) + \Theta(\log n);$$

$$T(1) = \theta(1)$$

Determine the asymptotic time complexity of $T(n)$.

A. $\Theta(\log n)$

B. $\Theta(n)$

C. $\Theta(n \log n)$

D. $\Theta(n^2)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

The tightest lower bound on the number of comparisons, in the Best case, for comparison-based sorting is of the order of

A. n

B. n^2

C. $n \cdot \log n$

D. $n \cdot \log^2 n$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider a weighted directed graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ and let X be a shortest path.

- S1: If we double the weight of every edge in the graph, then X will still be the shortest path.
- S2: If we square the weight of every edge in the graph, then X will still be the shortest path.

Which of the above statements are TRUE about X.

A. Both S1 and S2 are True

B. S1 is True and S2 is False

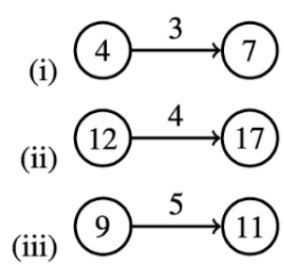
C. S1 is False and S2 is True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following directed edges with weights. What is the result of relaxing the following edges?



A. (i) - 7 (ii) - 17 (iii) - 11

B. (i) - 7 (ii) - 16 (iii) - 14

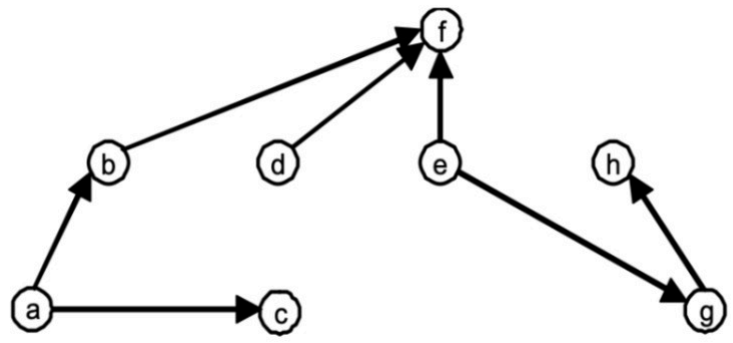
C. (i) - 7 (ii) - 16 (iii) - 11

D. (i) - 7 (ii) - 17 (iii) - 11

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Given The graph below:



Which of the following is not a topological ordering of vertices?

A. $abcdefgh$

B. $acbefdgh$

C. $acdbegfh$

D. $deacbfgh$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which of the following statements is/are True.

- $S1$: Every Directed Acyclic graph has exactly one topological ordering
- $S2$: Dijkstra's algorithm may not terminate if the graph contains negative-weight edges.

A. $S1$ is True and $S2$ is False

B. $S1$ is False and $S2$ is True

C. Both $S1$ and $S2$ are True

D. Both $S1$ and $S2$ are False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be a simple connected undirected graph with positive edge weights

- $S1$: For such a graph, the Bellman-Ford algorithm and Dijkstra's algorithm always compute the same shortest path weights from a given source but may return different shortest-path trees.
- $S2$: For such a graph, Dijkstra's algorithm always produces the unique shortest-path tree for any arbitrary source.

Which one of the following is CORRECT?

A. Both $S1$ and $S2$ are True

B. $S1$ is True and $S2$ is False

C. $S1$ is False and $S2$ is True

D. Both $S1$ and $S2$ are False

Your Answer: | **Correct Answer: B** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let n be a positive integer. Consider the two statements below:

- $S1$: If f(n) < g(n) for all n, then g(n) is ALWAYS ω(f(n)).
- $S2$: If f(n) > g(n) for all n, then f(n) is ALWAYS ω(g(n)).

Which of the following is the correct option?

A. $S1$ is correct but $S2$ is incorrect

B. $S1$ is incorrect but $S2$ is correct

C. Both $S1$ and $S2$ are correct

D. Both $S1$ and $S2$ are incorrect

Your Answer: | **Correct Answer: D** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

A system of difference constraints of the form $x_j - x_i \leq b_k$, can be represented by the help of directed graph $\mathrm{G}(\mathrm{V}, \mathrm{E})$, where $V = \{v_0, v_1, \ldots, v_n\}$ and

$$E = \left\{\left(v_i, v_j\right) : x_j - x_i \leq b_k\right\} \cup \left\{\left(v_0, v_1\right), \left(v_0, v_2\right), \ldots, \left(v_0, v_n\right)\right\}$$

Find the feasible solution for the following system of difference constraints:

$$\begin{aligned} x_1 - x_2 &\leq 1 \\ x_1 - x_3 &\leq -4 \\ x_2 - x_4 &\leq 7 \\ x_3 - x_2 &\leq 2 \\ x_4 - x_1 &\leq -1 \\ x_4 - x_3 &\leq 3 \end{aligned}$$

A. $x = \{2, 1, 5, 3\}$
B. $x = \{2, 1, 4, 3\}$
C. $x = \{5, 7, 9, 4\}$
D. $x = \{2, 4, 6, 1\}$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider a directed graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, where $\mathrm{V} = \{1, 2, \ldots, 100\}$ and $\mathrm{E} = \{(\mathrm{i}, \mathrm{j}) : \mathrm{i} < \mathrm{j} \leq \mathrm{i} + 2$, for all $\mathrm{i}, \mathrm{j} \in \mathrm{V}\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex 1. The number of connected components formed are 'A' and number vertices that will be discovered after vertex 100 is ' B ' then $6\ \mathrm{A} + \mathrm{B}$ _____

Your Answer: | Correct Answer: 55 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $\mathrm{lpm}(\mathrm{S}, \mathrm{i}, \mathrm{j})$ be a function that returns the length of the longest palindromic match in the String S of length n starting at index i and ending at index j (0 -based indexing). Consider the following recursive definition of $\mathrm{lpm}(\mathrm{S}, \mathrm{i}, \mathrm{j})$.

$$\mathrm{lpm}(S, i, j) = \begin{cases} 0 & \text{if } i > j \\ 1 & \text{if } i = j \\ 2 + \mathrm{lpm}(S, i+1, j-1) & \text{if } S[i] = S[j] \\ \max\{\mathrm{lpm}(S, i+1, j), \mathrm{lpm}(S, i, j-1)\} & \text{if } S[i] \neq S[j] \end{cases}$$

Which of the following statements is TRUE?

A. The function correctly computes the length of the longest palindromic substring in S .
B. The function correctly computes the length of the longest palindromic subsequence in S .
C. The function does not compute a correct value in any case.
D. The function may enter an infinite loop due to incorrect recursion.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

I. If $\mathrm{f(n)} = \mathrm{O(g(n))}$ and $\mathrm{g(n)} = \mathrm{O(h(n))}$, then $\mathrm{h(n)} = \Omega(\mathrm{f(n)})$
II. If $\mathrm{f(n)} = \mathrm{O(g(n))}$ and $\mathrm{g(n)} = \mathrm{O(f(n))}$ then $\mathrm{f(n)} = \mathrm{g(n)}$

Which one of the following is TRUE?

A. Both (I) and (II) is True

B. (I) is True and (II) is False

C. (I) is False and (II) is True

D. Both (I) and (II) is False

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Choice Type Award: 2 Penalty: 0.67 Algorithms

Which of the following recurrence relation claims is/are TRUE:

I. $\mathrm{T}(n) = 3\mathrm{T}(n/6) + cn$, for some positive constant c, and $\mathrm{T}(1) = 1$, then $\mathrm{T}(n) = O(n)$.
II. $\mathrm{T}(n) = \mathrm{T}(n/2) + \log n$, and $\mathrm{T}(1) = 1$, then $\mathrm{T}(n) = O(\log n)$.

A. Both (I) and (II) is True
B. (I) is True and (II) is False
C. (I) is False and (II) is True
D. Both (I) and (II) is False

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #21 Multiple Choice Type Award: 2 Penalty: 0.67 Algorithms

Consider a function that performs insertion sort on the array A of size n .

```
InsertionSort(A, n){
    COUNT_X = 0;
    For i = 0 to n {
        j = i - 1
        current = A[i]
        While array[j] > current {
            COUNT_X ++;
            shiftRight(array[j]);
            j++;
        }
        array[j] = current;
    }
    return COUNT_X;
}
```

Let:

- $\mathrm{COUNT_X}$: the number of shifts performed by the while loop
- $\mathrm{COUNT_Y}$: the number of comparisons made by the while loop

Which of the following statements is TRUE for some constant C?

A. For all n the function satisfies the relation $\mathrm{COUNT_X} \leq \mathrm{COUNT_Y} \leq C \cdot \mathrm{COUNT_X}$

B. There exist some n for which the function satisfies the relation $\mathrm{COUNT_Y} \leq \mathrm{COUNT_X} \leq C \cdot \mathrm{COUNT_Y}$

C. For all n the function satisfies the relation $\mathrm{COUNT_X} = \mathrm{COUNT_Y}$

D. There exist some n for which the function satisfies the relation $\mathrm{COUNT_X} = \mathrm{COUNT_Y}$

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #22 Multiple Choice Type Award: 2 Penalty: 0.67 Algorithms

An integer array A is $\boldsymbol{k}$-even-mixed if there are exactly k even integers in A, and the odd integers in A appear in sorted order. Given a k-even-mixed array A containing n distinct integers for $k = \lceil n/\lg n \rceil$, Which of the following is the minimum time complexity to sort A.

A. $\Theta(n)$

B. $\Theta(k.logn)$

C. $\Theta(n.logk)$

D. $\Theta(n.logn)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let G be a connected bipartite (graph having no odd length cycle) simple graph with distinct edge weights. Which of the following statements on MST (minimum spanning tree) are FALSE?

A. All MST of G contains lightest edge

B. All MST of G contains the third-lightest edge.

C. All MST of G contains the fourth-lightest edge.

D. No MST of G contains the heaviest edge.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following program:

```
x := 1
y := 0

  do while y > x:

    x := x/2

    y := y + 1

  end
```

For the above fragment, which of the following is a loop invariant (A loop invariant is a condition that remains true before and after each iteration of the loop)?

A. $x = \frac{1}{2^{y+1}}$

B. $x = \frac{x}{2^y}$

C. $x = \frac{1}{2^y}$

D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider a list of n positive integers a1, a2, . . . , an. Given an integer m, we query to determine if there is a prefix (a1, a2, ... , ai) of the list that sums to m. To solve such a query, we precompute all prefix sums and keep them in an array whose ith element is $\mathrm{a1} + \mathrm{a2} + \ldots +$ ai. The most efficient way of searching this list for integer m is to do a

A. Linear Search

B. Binary Search

C. Random Search (it randomly generates the position of element in O(1) time)

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Suppose we are in a casino playing the cards, after a series of losses, we have only 100 Rs remaining to play the next match, and we decide to bet the remaining money.

The game rules are as follows: We can win the 500 Rs. If we get a colour (a set of 4 cards where we have exactly 1 queen, 1 king, 1 Ace, 1 jack of any colour) and lose otherwise, but there's a catch: if we get all 4 aces together, then we get a chance to play 2 more times without betting any money.
Let:

- p1 = Probability of getting a "colour"
- p2 = Probability of getting all 4 Aces,
- p3 = Probability of losing the bet.

$\mathrm{E(x)}$ is the function that calculates the expected amount of money he wins or loses.
Which of the following recurrence relations about the function $\mathrm{E(x)}$ is correct?

A. $\mathrm{E(x)} = 500\mathrm{p1} + \mathrm{p2}(2 + \mathrm{E(x)}) - 100\mathrm{p3}$

B. $\mathrm{E(x)} = 500\mathrm{p1} + \mathrm{p2} * (2 * \mathrm{E(x)}) - 100\mathrm{p3}$

C. $\mathrm{E(x)} = 500\mathrm{p1} + \mathrm{p2}$
$\mathrm{E(x)} - 100\mathrm{p3}$

D. $\mathrm{E(x)} = 500\mathrm{p1} + 2\mathrm{p2}$
$\mathrm{E(x)} - 100(1 - \mathrm{p1} - \mathrm{p2})$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following statements about a graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$:

- $S1$: Given a connected unweighted graph G, if a vertex $\mathrm{v} \in \mathrm{V}$ is visited during level k of a Breadth-First Search (BFS) from a source vertex $s \in V$ then every path from s to v has length at least k.

- $S2$: A Depth-First Search (DFS) takes $\Theta\left(V^2\right)$ time on a graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ represented using an adjacency matrix.

- $S3$: Given an adjacency list representation of an undirected graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$, it takes $\mathrm{O}(\mathrm{V})$ time to compute the in-degree of every vertex.

Which of the above statements is/are TRUE?

A. $S1$ is True and $S3$ is True

B. $S1$ is True and $S2$ is False

C. $S1$ is True and $S2$ is True

D. All of the above

Your Answer: | **Correct Answer: C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ be an undirected connected graph. Suppose we run a Breadth-First Search (BFS) starting from an arbitrary vertex $v \in V$.

- The depth of the BFS tree is defined as the maximum distance (in number of edges) from v to any other vertex reachable from it. That is:

$$\mathrm{depth}(v) = \max_{u \in V} \delta(v, u)$$

- The diameter of the graph is the maximum shortest-path distance between any two vertices in the graph:

$$\mathrm{diameter}(G) = \max_{s,t \in V} \delta(s, t)$$

Where $\delta(\mathrm{x}, \mathrm{y})$ denotes the shortest-path distance between vertices x and y.
Which of the following statements is correct?

A. The depth of the BFS tree is always equal to the diameter of the graph.
B. The depth of the BFS tree from any vertex is always greater than or equal to the diameter.
C. The depth of the BFS tree from any arbitrary vertex is always less than or equal to the diameter.
D. The depth of the BFS tree from any arbitrary vertex is always equal to the eccentricity(maximum distance from v to any other vertex) of the vertex with maximum degree.

Your Answer: | **Correct Answer: C** | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $\mathrm{T}(\mathrm{n}) = 4\ \mathrm{T}(\mathrm{n}/2) + \theta(\mathrm{n})$, where "$O$" is big-oh. What will be the asymptotic bound on $\mathrm{T}(\mathrm{n})$?

A. $O(n)$

B. $O(nlogn)$

C. $O\ (n\hat{}2)$

D. $O\ (n\hat{}3)$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose that A is an array of n elements. Each element of A has some flavour; you cannot tell the flavours apart, but you know that one flavour is in the strict majority. A flavour is said to be in strict majority if more than n/2 elements are the same. You have access to a genie called isMajority so that isMajority returns TRUE if x is an element of A with majority flavour and FALSE otherwise.

```
def findMajority(A):

  n = len(A)  # assume that n is a power of 2.

  if n == 1:

    return A[0]

  a = findMajority(A[:n//2])

  b = findMajority(A[n//2:])

  if isMajority(A, a):

    return a

  else:

    return b
```

Suppose that $isMajority$ runs in time $\theta(\sqrt{n})$ on an array of length n. What is the running time of $findMajority$?

A. $\theta(\sqrt{n})$

B. $\theta(n)$

C. $\theta(n.logn)$

D. $\theta(n2)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Copyright & Stuff